

KENNETH W. BINGHAM

AI Systems Architect · Senior Software Engineer · Full-Stack Platform Owner

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Portfolio: butterflyfx.us · kensgames.com · ttlrecall.com · dimensionsos.com

SUMMARY

Ten-plus-year senior engineer who designs and ships AI systems whose behavior is **provable**, not hoped-for. I build directive frameworks that hold against drift, agent loops that route by logic gates and minimal-surface decision boundaries, and full-stack platforms I own end-to-end — from manifold compiler to nginx config. Recent focus: AI systems architecture, deterministic agent design, and dimensional / manifold programming.

WHAT I OPTIMIZE FOR

- **Maximum quality per dollar of AI spend.** Route by truth-table constraints, decision-tree shortcuts, and a Schwarz-D minimal-surface boundary so frontier credits go only where frontier reasoning is required.
- **Determinism over fluency.** Schema-validated outputs, retry budgets, verifier ensembles — LLMs never return free text to the next stage.
- **Ancestry-preserving systems.** No destructive updates; every state is traceable through its lineage on the underlying manifold.
- **Sovereign over rented.** I own the VPS, the deploy pipeline, the data model, the directive doctrine. Vendor portability is a first-class requirement.

CORE TECHNICAL SKILLS

AI & Agents	Directive engineering · agent loops · evaluation harnesses · tool-use design · RAG · AWS Bedrock Knowledge Bases · vector search (OpenSearch, Pinecone, Milvus, local bge) · multi-model routing (Claude, GPT, Gemini, Llama, Copilot) · non-drifting behavior frameworks
Languages	Java 8/11/17 · C# / .NET 6 / ASP.NET Core · Python 3 · TypeScript / JavaScript (ES2022+) · SQL · Bash / PowerShell
Backend	Spring Boot · Quarkus · REST · microservices · HL7 / FHIR · ETL pipelines · deterministic state engines · SQLite-backed record layers
Frontend / 3D	Angular · React · Three.js / WebGL / WebGPU · interactive 3D · minimal-surface visualization · SVG instrumentation
Cloud / DevOps	GitHub Actions · Azure Pipelines · Docker · Kubernetes · OpenShift · Linux (Ubuntu) admin · NGINX · PM2 · SSL / domain routing · VPS hosting end-to-end
Data	SQL Server · Oracle · MySQL · Postgres · SQLite · MarkLogic · document & vector stores · schema design · indexing · query optimization
Security / Compliance	Mission-critical DoD systems · healthcare HL7/FHIR · financial-data parsing · secure authentication workflows · on-machine LLM routing for PII

INDEPENDENT PLATFORMS — SOLE ARCHITECT & DEVELOPER

KensGames.com — full 3D game portal with deterministic simulation. *Manifold compiler (Python) · Three.js engine · Node/Express auth + lobby · SQLite records layer · PM2 + nginx on VPS · GitHub Actions deploy pipeline.* Built eight games (fasttrack, 4DTicTacToe, brickbreaker3d, starfighter, cubic3d, assemble, ...) on a single substrate-driven rule engine. Each game is a manifold projection — no per-game procedural code.

ButterflyFX.us — portfolio + research surface for dimensional programming. *Static site · Three.js spirals · Schwarz-D routing module (live demo) · doctrine documents.* Hosts the Ultimate AI Directive, the Schwarz-D AI routing system (interactive demo with truth-table constraints, decision tree, Tiresias manifold, surface-boundary tier selection, determination graph plan), and a downloadable routing policy.

TTLRecall.com — verified scenario generation platform. *AI-accelerated workflow · two complete production platforms built and deployed in ~30 minutes.* Demonstrates how directive-driven AI development collapses traditional dev cycles without giving up architectural rigor.

DimensionSOS.com — technical hub for stateless, deterministic, AI-aligned system design. Reference architecture and writing surface for the dimensional-programming methodology.

PROFESSIONAL EXPERIENCE

Software Developer — DTSI

Apr 2024 – Present · Hill Air Force Base

- Modernize mission-critical Air Force systems in Java, .NET, and SQL Server under secure DevOps practices; reduce regression risk by introducing deterministic test harnesses around legacy workflows.
- Operate inside a DoD-cleared environment with strict change-control and supply-chain constraints.

Software Developer — Humana

Feb 2022 – Feb 2024 · Remote

- Built and shipped features across a large-scale healthcare platform: .NET 6 services, Azure Pipelines CI/CD, HL7 / FHIR data flows, SQL query tuning.
- Stood up Python-based ETL plus Linux administration that automated previously manual deployment and data-prep cycles.

Software Developer — Systems Implementers

Apr 2021 – Dec 2021 · Hill Air Force Base

- Built Java / Spring Boot services for aircraft maintenance and asset-tracking workflows; integrated with existing military command systems.

Software Developer — Gwinnett County School District

Feb 2020 – Aug 2020 · GA

- Authored automated test frameworks in Java + Selenium WebDriver; expanded QA coverage and shortened release-validation cycles.

Software Developer — Norfolk Southern Railroad

Aug 2019 – Jan 2020 · Atlanta, GA

- Improved locomotive-optimization software (Java, JavaScript); handled real-time data processing and incremental legacy modernization.

Software Developer — Finicity (via STG)

Nov 2017 – Aug 2018 · Murray, UT

- Built Java-based financial-data extraction tooling; hardened API integrations, parsing pipelines, and secure handling of sensitive consumer data.

Java Programmer — Wex

Feb 2017 – Sep 2017 · Ogden, UT

- Delivered enterprise web portals and document-management workflows on Oracle, Spring, and enterprise messaging.

Java Programmer — CUProdigy

Apr 2015 – Feb 2017 · UT

- Developed secure banking applications in Java, Spring, Hibernate, and MySQL; strengthened authentication workflows and backend reliability across the credit-union platform.

EDUCATION

B.S., Computer Science & Software Engineering

SELECTED WRITING & OPEN SOURCE

- **Substack (@kbingh)** — AI architecture, dimensional computing, next-generation system design.
- **github.com/kenbin64** — KensGames manifold compiler, ButterflyFX site + router, scenario generation tooling, and supporting modules.
- **The Ultimate AI Directive** (butterflyfx.us/directive.html) — nine-section governing framework for manifold-based AI behavior.
- **Schwarz-D AI Routing** (butterflyfx.us/routing.html) — five-layer model-routing system with live demo and downloadable policy.